Run report: Gröbner engines against msolve, Singular and PolyBoRi, run-20261011T1806Z

Observations only. Interpretation: `../../REPORT.pdf`.

What ran

- **Runner:** `research/gb_cas_comparison_20261011/run.sh <bin> <this dir> 3 600 120`, with `POLYBORI_PY` set. Binaries: release builds of `examples/gb_cas_export.rs` and `examples/f6_ic_probe.rs` from commit `6bdf93f4b` plus the new exporter (the engine sources equal PR #1671's).
- **External tools** (`raw/versions.txt`): msolve 0.6.5-1build2 and Singular 4.3.2-p10 (Ubuntu packages); PolyBoRi through `passagemath-brial` 10.8.12 (Python 3.13). Magma: **not run**. It is not installed and needs a licence; its scripts are in `inputs-fp/*.magma` and `inputs-bool-n17/*.magma`.
- **Host:** Claude Code cloud container, Intel Xeon @ 2.10 GHz, 4 vCPU, shared, no pinning (isolation L0).
- **F_p workload:** the 13 systems of `examples/f4_fp_bench.rs` (same generator and seeds; basis fingerprints equal the bench's). Ours: `f4_fp::f4`, grevlex, the bench's degree bound, median of 3 repetitions per round, with `RAYON_NUM_THREADS` 1 and 4. msolve: `-g 2` (reduced GB), `-t 1` and `-t 4`, wall time of the process (and its own `overall(elapsed)`, 10 ms resolution). Singular: `std` under `dp` with `redSB`, timed by `rtimer` inside Singular. Three rounds, interleaved.
- **Boolean workload:** the 8 root decomposition systems of `f6_ic_probe --a 1 --n 17 --dim 6 --targets 8` (Koblitz K_1 over $\mathbb{F}_{2^{17}}$, $m = 3$, a 63-point factor base; 35 Boolean variables and 34 cubics each, from `KIC_F4_NODE_DUMP`). Ours: the probe's per-target decision time, best of 3 repetitions, for inherited F4 (`f4`) and F6-IC with the pair index (`f6p`). External: one full reduced Gröbner basis with the field equations (msolve `-t 1`, Singular `std`), or in the Boolean ring (PolyBoRi `groebner_basis`). Time limits: 600 s (msolve, PolyBoRi) and 120 s (Singular).
- **Requested and completed:** all F_p runs completed. Boolean: msolve and PolyBoRi completed 8 of 8 each; Singular completed 0 of 8 (all hit the 120 s limit, exit 124).

F_p: reduced Gröbner bases (median ms of 3 rounds)

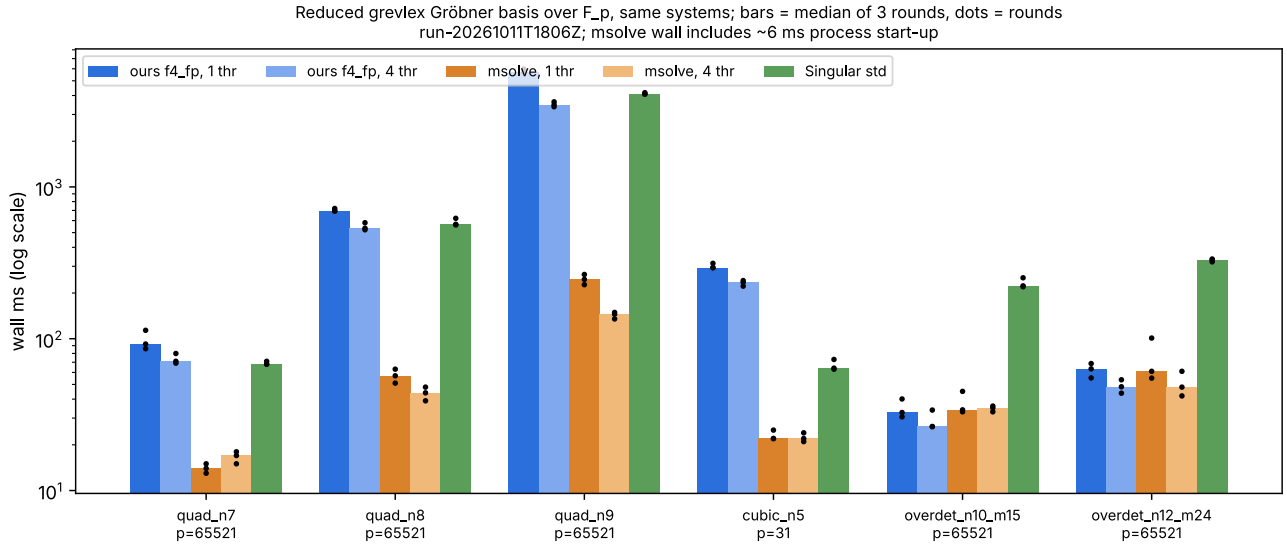


Figure 1: F_p cases with at least 10 ms of work, run-20261011T1806Z.

case	ours 1 thr	ours 4 thr	msolve 1 thr	msolve 4 thr	Singular	same basis
quad_n4_p31	0.6	0.6	7	8	0	yes
quad_n5_p31	2.8	2.9	7	8	1	yes
quad_n6_p65521	17.0	13.7	8	10	9	yes
quad_n7_p65521	92.3	71.1	14	17	68	yes
cubic_n3_p29	1.4	1.2	6	9	0	yes
cubic_n4_p31	18.8	14.4	9	10	3	yes
quart_n3_p31	8.0	6.4	7	10	1	yes
overdet_n6_m9_p101	1.1	1.2	7	10	1	yes
quad_n8_p65521	694.7	534.1	57	44	567	yes
quad_n9_p65521	5693.2	3479.0	246	145	4106	yes

cubic_n5_p31	294.2	234.8	22	22	64	yes
overdet_n10_m15_p65521	32.7	26.4	34	35	223	yes
overdet_n12_m24_p65521	63.3	48.3	61	48	331	yes

All measured. “Same basis”: in every round, Singular found our basis to be a reduced Gröbner basis of the same ideal as its own (`same_ideal 1`, `same_size 1`: equal size, mutual normal forms zero, equal leading terms). msolve’s wall time includes process start-up and parsing (about 6–7 ms at the smallest cases).

Boolean: deciding the n = 17 decomposition systems (ms)

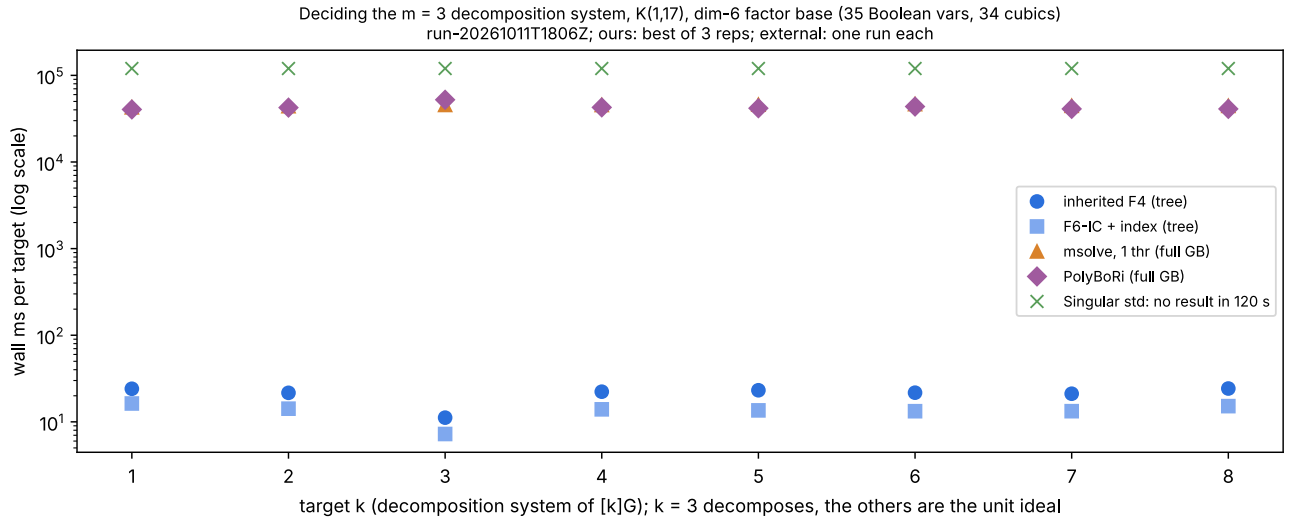


Figure 2: Per-target decision time, run-20261011T1806Z. Singular’s points are the 120 s limit, not a result.

target	ours f4	ours f6p	msolve	PolyBoRi	Singular	verdict (all that finished)
1	24.1	16.3	42 741	40 421	> 120 000	unit ideal
2	21.6	14.2	44 061	42 448	> 120 000	unit ideal
3	11.2	7.2	45 517	52 296	> 120 000	decomposes
4	22.3	13.9	45 459	42 692	> 120 000	unit ideal
5	23.2	13.6	45 845	41 753	> 120 000	unit ideal
6	21.7	13.3	46 527	43 659	> 120 000	unit ideal
7	21.1	13.3	44 485	41 012	> 120 000	unit ideal
8	24.3	15.2	44 436	41 005	> 120 000	unit ideal

Verdicts agree: ours (`hit`), msolve (`[1]` or not) and PolyBoRi (`unit_ideal`) give the same answer on all 8 targets. Ours decides by splitting variables and degree-bounded inherited Macaulay bases; the external tools compute the full reduced basis of the ideal. The two are different computations that answer the same question.

Records

`raw/fp_ours_t{1,4}.round{1,2,3}.jsonl`, `raw/fp_external.round{1,2,3}.tsv`, `raw/*.msolve_t*.out.gz` (msolve’s bases), `raw/bool_n17_ours.jsonl`, `raw/bool_n17_external.tsv`, `raw/bool_n17_roots.jsonl`, `raw/versions.txt`; inputs in `inputs-fp/` and `inputs-bool-n17/` (`.ms` msolve, `.sing` Singular, `.magma` Magma). Graph source: `plot.py`, which reads `raw/` only.

Missing

Magma timings: not run (no Magma here). No isolated host and no interval beyond the three F_p rounds; one external run per Boolean root. The n = 23 decomposition systems (47 variables) were not run externally.